ARTPARK

ಕರ್ನಾಟಕ ಸರ್ಕಾರ

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

Telephone: 080 – 23341652, 23348848 || Email: spp@kscst.org.in
Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 49th SERIES OF STUDENT PROJECT PROGRAMME

(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form.

https://forms.gle/iXiR81MyoLWn871x7

1.	**Name of the College:** **K.S. INSTITUTE OF TECHNOLOGY**
2.	**Project Title:** **AI-Based Blood Group Prediction and Emergency Health Profile System**
3.	**Branch:** **Computer Science and Design**
4.	**Theme (as per KSCST poster):** **Leveraging Artificial Intelligence to enable rapid, contactless blood group identification and instant retrieval of emergency health data, supporting faster medical decision-making, improved patient safety, and enhanced emergency response**
5.	**Name(s) of project guide(s):** **1. Name: Mamatha .C** **Email id: mamatha.c@ksit.edu.in** **Contact No: 8105406179**
6.	**Name of Team Members (Strictly not more than four students in a batch):** *(Type names in Capital Letters as provided in your college)* (Please paste the latest passport size photograph adjacent to your respective names) **Name : P ANUDEEP** **USN No : 1KS22CGO04** **Email id : anudeep5065@gmail.com** **Mobile No : 7396555113**

Name: E DURGA MAITRI
USN No.: 1KS22CG014
Email id: durgamaitri2004@gmail.com
Mobile No.:8217597464

Name: HARSHITHA P
USN No.: 1KS22CG016
Email id: harshithap0204@gmail.com
Mobile No.: 7975735390

Name: MADHUSRI PM
USN No.: 1KS22CG025
Email id: madhusripm@gmail,com
Mobile No.: 7975411813

7. Team Leader of the Project:

Name: P Anudeep
USN No.:1KS22CG004
Email id: anudeep5065@gmail.com
Mobile No.: 7396555113

8. Processing Fee Details (Through Online Payment only):
(processing fee of Rs. 1180/-)

Payment completed through UPI: Phone Pay

UPI no: 7396555113

Transaction ID:

9. Date of commencement of the Project:
24th March 2025

10. Probable date of completion of the project:
December 2nd 2025

11. Timeline Structure

Task	Start Date	End Date	Duration	Milestone achievable
Project Initiation	24 th March 2025	22nd Apri 2025	1 month	Critical thinking and Project planning
Research Phase	23rd April	24th May	1 Month	Gathering Data

	2025	2025		and Resources
Development Phase	25th May 2025	28th September	4 Months	Training Model and Coding
Testing Phase	29th September 2025	November 6th 2025	1 Month	Testing Data with other Foreign Samples
Submission of PPT and Abstract to KSCST for conducting mid-term evaluation (Upload to Google Form)	Middle of March – 2nd Week of April 2026			
Final Review				
Project Submission				
Submission of Project Completion Report to KSCST (Upload to Google Form)	April / May 2026			

12. **Scope / Objectives of the project:**

- To develop an intelligent system that predicts a person's blood group using fingerprint patterns and maintains an emergency medical profile to support quick healthcare responses.
- To predict the blood group of an individual using AI and fingerprint image analysis.
- To develop an emergency health profile system that stores and retrieves patient data instantly.
- To reduce diagnosis time during emergencies by providing quick access to blood group and medical details.
- To integrate AI and biometric technologies for reliable and automated health identification.
- To enhance healthcare efficiency through digitalization and intelligent prediction systems.
- To create a portable and cost-effective system suitable for both urban and rural healthcare centers.
- To minimize human error in blood group identification and data recording.
- To improve patient safety and treatment outcomes through real-time, accurate medical information.

13. **Methodology:**

1. Problem Identification & Requirement Analysis
   - Identify the need for a quick and AI-driven blood group detection system.
   - Collect requirements from healthcare professionals and emergency response teams.
   - Define key modules: Blood group prediction, health profile storage, and data

access.

2. Data Collection
    - Gather fingerprint images and corresponding blood group data for training the AI model.
    - Use open-source biometric datasets or collect real samples under supervision.
    - Preprocess data by removing noise and normalizing image sizes.
3. Image Preprocessing
    - Apply image enhancement techniques (filtering, edge detection, segmentation) to improve fingerprint clarity.
    - Convert images into grayscale and extract key features such as ridges, bifurcations, and patterns.
    - Store preprocessed images in a structured database.
4. Feature Extraction
    - Use image processing algorithms (like Gabor filters or Minutiae extraction) to identify unique fingerprint features.
    - Generate numerical representations (feature vectors) for AI model input.
5. AI Model Training and Prediction
    - Train a machine learning or deep learning model (e.g., CNN – Convolutional Neural Network) using labeled fingerprint data.
    - The model learns to associate fingerprint patterns with specific blood groups.
    - Evaluate accuracy using training and testing datasets.
6. System Design and Development
    - Design the system architecture including:
        - Frontend: User interface for data input and display.
        - Backend: AI model integration and database management.
        - Database: Stores fingerprint data, health records, and user profiles.
    - Develop using suitable tools (Python, TensorFlow/Keras, OpenCV, MySQL, Flask/Django).
7. Emergency Health Profile Integration
    - Create a module where users can register health details (allergies, diseases, medications, etc.).
    - Link the health profile to the user's fingerprint ID for instant retrieval.
8. Testing and Validation
    - Perform unit testing, system testing, and user acceptance testing to ensure reliability.
    - Validate AI prediction accuracy and system performance in real-time scenarios.
9. Deployment and Implementation
    - Deploy the system on a cloud or local server for healthcare use.

	○ Train healthcare staff to use the system efficiently during emergencies. 10. Maintenance and Future Enhancement ○ Regularly update the database and AI model with new data. ○ Enhance the system to include additional biometric or health parameters (e.g., heart rate, oxygen level).
14.	**Expected Outcome of the project:** **Accurate Blood Group Prediction** • The system will successfully predict a user's **blood group** by analyzing their **fingerprint pattern** using AI algorithms. • The prediction accuracy is expected to be **above 90%** with a properly trained model. **Automated Identification System** • Users can simply **scan their fingerprint**, and the system will automatically identify their blood group and retrieve stored health information. **Digital Emergency Health Profile** • A **personalized health profile** will be created and linked with the user's fingerprint. • It will contain essential details like: ○ Name and contact information ○ Blood group ○ Medical history (e.g., allergies, chronic diseases) ○ Emergency contact number **Quick Access in Emergencies** • In case of accidents or critical medical situations, doctors or paramedics can **instantly access** a patient's health and blood group data through fingerprint scanning. **User-Friendly Interface** • The system will have a **simple graphical interface** for users and healthcare professionals to interact easily. • Options for registration, data update, and emergency mode access will be available. **Database and AI Integration** • A **centralized database** will securely store fingerprint data, AI-predicted results, and health records. • The **AI model** will be integrated to provide real-time predictions during new scans. **Improved Healthcare Efficiency** • The system will **reduce manual testing time** and improve response time in emergencies. • It ensures **accuracy, reliability, and fast diagnosis**, enhancing overall healthcare service quality. **Scalable and Extendable System** • The system can be extended to include additional health analytics like **disease risk prediction** or **vital monitoring** in the future.

15.	**Is the project proposed relevant to the Industry / Society or Institution?** **Yes / No: No** **If yes, please provide details of the industry / institution and contact details:** (**Note:** Preference will be given to those projects relevant to the industry / institution. Hence be specific in giving detailed information). Is the industry extending support - technology / funds / use the final product, please specify.
16.	**Can the product or process to be developed in the project be taken up for filing a Patent?** **Yes / No: NO** **Prior Art search done?** **Yes/No: NO** **Note:** If your answer is "Yes", you may contact Patent Information Centre of KSCST. For more details, email: pic@kscst.org.in
17.	**Budget details (break-up details should be given):** Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee.

Budget	Amount
a) Materials / Consumables (Please specify)	3000
b) Labor (Describe)	0.00
c) Travel (Describe)	300
e) Miscellaneous (Please specify)	650
Total	3950.00

18.	**SPP Coordinator (Identified by the college):** **Note:** To be identified by the principal of the institution. The project proposals must be submitted to KSCST through SPP coordinator designated by the principal. **Name: Prof. / Dr. Bharathi Gururaj** **Email id: bharathigururaj@ksit.edu.in** **Contact No.: 9980095899**

Dr. Bharathi Gururaj

(Name & Signature of
SPP Coordinator with Seal)
Email id: bharathigururaj@ksit.edu.in
Contact No.: 9980095899

12/12

Name and signature of the
Project Guide with Seal:
Email id: Mamatha.c@ksit.edu.in
Contact No.: 8105406179

DECLARATION

(From Project Students)

(To scan this page and enclose in the project proposal)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: **AI-Based Blood Group Prediction and Emergency Health Profile System.** College: **K.S INSTITUTE OF TECHNOLOGY** are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.		**Signature with date**
1.	P ANUDEEP	P. Anudeep 12/12/25
2.	E DURGA MAITRI	Maitri 12/12/25
3.	HARSHITHA P	Harshitha P 12/12/25
4.	MADHUSRI PM	Madhusri. P.M 12/12/25

BG
Dr. Bharathi Gururaj

(Name & Signature of SPP Coordinator with Seal)

Email id: bharathigururaj@ksit.edu.in

Contact No.: 9980095899

12/12

Name and signature of the Project Guide with Seal:

Email id: mamatha.c@ksit.edu.in

Contact No.: 8105406179

Kammavari Sangham (R) 1952
ಕೆ.ಎಸ್. ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ
K. S. INSTITUTE OF TECHNOLOGY
An Autonomous Institution under VTU
Approved by AICTE, New Delhi, Recognized by Govt. of Karnataka, Accredited by NBA (CSE & ECE) and NAAC with A+ Grade
#14, Raghuvanahalli, Kanakapura Road, Bengaluru - 560 109.
Tel: 080-28427163/164. website : www.ksit.ac.in, E-mail ID : principal@ksit.edu.in / principal.ksit@gmail.com

ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that **1) Mr. P Anudeep, 2) Ms. E Durga Maitri 3) Ms. Harshitha P, 4) Ms. Madhusri PM** are bonafide student(s) of **Department of Computer Science**, in the degree program of our institution. If the project proposal submitted by these students under the 49th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.

(Name & Signature of Project Guide with Seal)	**(Name & Signature of SPP Coordinator with Seal)**	**(Signature of HoD with Seal)**
Email id: Mamatha.c@ksit.edu.in	**Email id: bharathigururaj@ksit.edu.in**	**Email id: deepasr@ksit.edu.in**
Contact No.: 9980095899	**Contact No.: 8105406179**	**Contact No.: 9880101980**

(Signature of the Principal with Seal)

Email id: principal@ksit.edu.in

Contact No.: 9606064187

PRINCIPAL
K.S. INSTITUTE OF TECHNOLOGY
BENGALURU - 560 109.

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note**: Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848, email: finance@kscst.org.in

1. TITLE OF THE PROJECT	:	**AI-Based Blood Group Prediction and Emergency Health Profile System**
2. NAME OF THE TEAM LEADER	:	**P ANUDEEP**
3. EMAIL ID	:	**anudeep5065@gmail.com**
4. CONTACT MOBILE NO.	:	**7396555113**

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	858071963279
6. TRANSACTION ID	:	T2512121230433941108067
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	Anudeep Poola
8. NAME OF THE BANK	:	State Bank Of India
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	12/12/2025
11. TIME	:	12:30 PM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	UPI (PhonePay)

P. Anudeep

(Name & Signature of the team leader)

12/12

(Name & Signature of Project Guide / SPP Co-ordinator / HOD with Seal)

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

49th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in

केनरा बैंक Canara Bank

भारत सरकार का उपक्रम A Government of India Undertaking

सिंडिकेट Syndicate

KARNATAKA STATE COUNCIL FOR SCIENCE

0683201000024@cnrb

BHIM BHARAT INTERFACE FOR MONEY | UPI UNIFIED PAYMENTS INTERFACE

G20

भारत 2023 INDIA

वसुधैव कुटुम्बकम्

ONE EARTH • ONE FAMILY • ONE FUTURE

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in